

EXAM 7 – SPRING 2014

16. (2 points)

Given the following information for a publicly traded corporation:

Market risk premium:	6.0%
Return on 90-day T-Bills:	2.5%
Beta for this corporation:	1.25

The corporation's dividend payments have been projected as follows:

Calendar <u>Year</u>	Dividend <u>Payment</u>
2014	\$450,000
2015	500,000
2016	675,000

After 2016, the corporation's dividend payments are expected to grow at 5% a year.

Estimate the value of this corporation as of December 31, 2013, using the dividend discount model.

CONTINUED ON NEXT PAGE

SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 16																					
TOTAL POINT VALUE: 2	LEARNING OBJECTIVE: B2: Value the equity of a P&C insurer based on its expected future dividends, its free cash flow to equity, or its expected abnormal earnings																				
SAMPLE ANSWERS																					
<i>Sample Answer 1</i>																					
$g = 0.5$ $r_f = 0.25$ $k = r_f + \beta(E(r_m) - r_f)$ $= 0.25 + 1.25 \times 0.06$ $= 0.1$ $\text{Terminal value} = \frac{2016 \text{ dividend} \times (1 + g)}{k - g} = \frac{675,000 \times 1.05}{0.1 - 0.05} = 14,175,000$ $\text{Total Value} = \frac{450,000}{1 + k} + \frac{500,000}{(1 + k)^2} + \frac{675,000}{(1 + k)^3} + \frac{\text{Terminal Value}}{(1 + k)^3}$ $= 409,091 + 413,223 + 507,137 + 10,649,887$ $= 11,979,338$																					
<i>Sample Answer 2</i>																					
$k = r_f + \beta(r_m - r_f) = 0.025 + 1.25(0.06) = 0.1$ $g = 0.05$ <table><tr><th>CY</th><th>DIV</th><th>PV Factor</th><th>PV(DIV)</th></tr><tr><td>2014</td><td>450,000</td><td>1/1.1=0.9091</td><td>409,091 = 450,000 × 0.9091</td></tr><tr><td>2015</td><td>500,000</td><td>1/1.1²=0.8264</td><td>413,223</td></tr><tr><td>2016</td><td>675,000</td><td>0.7513</td><td>507,137</td></tr><tr><td>Total</td><td></td><td></td><td>1,329,451</td></tr></table> $TV = \frac{DIV_{16}(1 + g)}{k - g} = \frac{675,000(1 + 0.05)}{0.1 - 0.05} = 14,175,000$ $PV(TV) = \frac{14,175,000}{1.1^3} = 10,649,887$ $\text{Total Value} = 10,649,887 + 1,329,451 = 11,979,338$		CY	DIV	PV Factor	PV(DIV)	2014	450,000	1/1.1=0.9091	409,091 = 450,000 × 0.9091	2015	500,000	1/1.1 ² =0.8264	413,223	2016	675,000	0.7513	507,137	Total			1,329,451
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EXAMINER'S REPORT																					
This question is a commonly asked question that stems directly from Learning Objective B2, Knowledge Statement (a) – “Dividend Discount Model.” Approximately 82% of candidates received full credit, and for the most part provided model solutions very similar to the items listed in the textbook.																					
While the large majority scored perfect, the minor mistakes that we did see occur were: rounding errors, no labeling of \$000's, and calculation errors.																					

SAMPLE ANSWERS AND EXAMINER'S REPORT

In addition, there was a common error where candidates calculated an incorrect discount rate using the CAPM formula of 6.875% instead of the correct 10.0%. In nearly all of these cases the remainder of the solution was correct and the candidates received only a minor deduction for the initial error.